La Scuola d'Italia / Dr. Huson / IB Math 12 HL
14 October 2026 Markscheme

12.6 Quiz: Complex Numbers — Markscheme

Calculator allowed. Where a part asks for an exact form, a GDC decimal earns the method marks shown but not the final A1.

1. [Maximum mark: 6]

   $z_1 = 3 + 4i, \quad z_2 = 1 - 2i.$

   (a) $z_1 z_2^* = (3+4i)(1+2i) = 3 + 6i + 4i + 8i^2$ **M1**

   $= -5 + 10i$ **A1**

   M1 for using $z_2^* = 1 + 2i$ and expanding. Using z_2 in place of its conjugate, giving $11 - 2i$, earns M0A0.

   (b) $\dfrac{3+4i}{1-2i} \cdot \dfrac{1+2i}{1+2i} = \dfrac{-5+10i}{5}$ **M1**

   $= -1 + 2i$ **A1**

   M1 for multiplying numerator and denominator by $1 + 2i$ (the numerator is the product from (a)), or for a GDC entry shown. FT: award M1 for dividing the candidate's answer to (a) by $|z_2|^2 = 5$.

   (c) $\left|\dfrac{z_1}{z_2}\right| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}\ (= 2.24)$ **A1**

   $\arg \dfrac{z_1}{z_2} = \pi - \arctan 2 = 2.0344\ldots = 2.03$ **A1**

   Accept $\frac{|z_1|}{|z_2|} = \frac{5}{\sqrt{5}}$ for the modulus. For the argument accept 2.03 radians or $117°$ ($116.565\ldots°$). $\arctan(-2) = -1.11$ (the fourth-quadrant angle) earns A0: $-1 + 2i$ lies in the second quadrant. FT from the candidate's quotient in (b).

2. [Maximum mark: 7]

$z = -\sqrt{3} + i, \quad w = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$

(a) $|z| = \sqrt{3+1} = 2$ **A1**

$\arg z = \pi - \dfrac{\pi}{6} = \dfrac{5\pi}{6}$, so $z = 2e^{i\frac{5\pi}{6}}$ **A1**

Second quadrant: $\arctan\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$ *alone earns A0 for the argument. Accept* 2.62 *for the argument.*

(b) $zw = 2 \times 3\,\text{cis}\left(\dfrac{5\pi}{6} + \dfrac{\pi}{4}\right) = 6\,\text{cis}\,\dfrac{13\pi}{12}$ **M1**

$= 6\left(\cos\left(-\dfrac{11\pi}{12}\right) + i\sin\left(-\dfrac{11\pi}{12}\right)\right)$ **A1**

M1 for multiplying the moduli and adding the arguments. A1 requires the argument in the stated range; $\frac{13\pi}{12}$ *left unadjusted earns M1A0. Accept* -2.88 *for the argument. FT from (a).*

(c) $zw = -5.80 - 1.55i$ **A1**

Exact form: $zw = -\frac{3}{2}(\sqrt{6}+\sqrt{2}) - \frac{3}{2}(\sqrt{6}-\sqrt{2})\,i = -5.7955\ldots - 1.5529\ldots i$. *Accept either the exact or the 3 s.f. form, from (b) or from multiplying the Cartesian forms directly.*

(d) A rotation about the origin through $\dfrac{\pi}{4}$ anticlockwise **A1**

and an enlargement, centre the origin, scale factor 3 **A1**

The two marks are independent. The rotation needs the angle and the direction; "turned by $45°$ *anticlockwise" is fine. "Stretched by* 3 *" or "three times as far from the origin" earns the second A1.*

3. [Maximum mark: 13]

$z^3 + az^2 + bz + 26 = 0, \quad a, b \in \mathbb{R}$, root $3 + 2i$.

(a) $3 - 2i$ **A1**

The coefficients are real, so non-real roots occur in conjugate pairs. **R1**

R1 requires the reference to real coefficients (or to a, b and 26 being real).

(b) Product of the roots $= -26$ **M1**

$(3 + 2i)(3 - 2i)\,\gamma = 13\gamma$ **A1**

$13\gamma = -26$, so $\gamma = -2$ **A1**

M1 for $(-1)^3 \times 26$ or for $\alpha\beta\gamma = -26$, stated or used. The product $+26$ leads to $\gamma = 2$: M0A1A0. FT from the candidate's second root in (a).

(c) *Method 1 (sum and pairwise products):*

Sum of the roots $= (3 + 2i) + (3 - 2i) + (-2) = 4 = -a$ **M1**

$a = -4$ **A1**

Sum of pairwise products $= 13 + (-2)(3 + 2i) + (-2)(3 - 2i) = 13 - 12 = 1$ **M1**

$b = 1$ **A1**

Method 2 (expanding the factors):

$(z + 2)(z^2 - 6z + 13)$ **M1**

$= z^3 - 4z^2 + z + 26$ **M1**

$a = -4, \quad b = 1$ **A1A1**

Substituting $z = 3 + 2i$ and equating real and imaginary parts is also valid (M1 for the substitution, M1 for equating both parts). FT from the candidate's roots.

(d) A $(3, 2)$, B $(3, -2)$, C $(-2, 0)$

AC $= |(3 + 2i) - (-2)| = |5 + 2i| = \sqrt{29}$ and BC $= |5 - 2i| = \sqrt{29}$ **M1**

AC = BC, so triangle ABC is isosceles. **A1 AG**

M1 for both lengths found (as moduli or by Pythagoras). Accept the symmetry argument: A and B are reflections of each other in the real axis, on which C lies, so CA = CB.

(e) Base AB $= 4$, height $= 3 - (-2) = 5$ **M1**

Area $= \frac{1}{2} \times 4 \times 5 = 10$ **A1**

FT from the candidate's real root.

4. [Maximum mark: 14]

(a) Let $P(n)$ be $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

$n = 1$: $(\cos\theta + i\sin\theta)^1 = \cos\theta + i\sin\theta$, so $P(1)$ is true. **A1**

Assume $P(k)$ is true for some $k \in \mathbb{Z}^+$: $(\cos\theta + i\sin\theta)^k = \cos k\theta + i\sin k\theta$. **M1**

$(\cos\theta + i\sin\theta)^{k+1} = (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$ **M1**

$= \cos k\theta\cos\theta - \sin k\theta\sin\theta + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta)$ **A1**

$= \cos(k+1)\theta + i\sin(k+1)\theta$ **A1**

$P(1)$ is true, and $P(k)$ true implies $P(k+1)$ true; so $P(n)$ is true for all $n \in \mathbb{Z}^+$. **R1**

The assumption must be stated as an assumption ("assume", "suppose", "let"); "$P(k)$ is true" stated as fact earns M0. The second A1 needs the compound angle identities applied to both parts. R1 only if the first four marks are earned and the conclusion names both the base case and the inductive step.

(b) $(\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta$ **M1**

Write $c = \cos\theta$, $s = \sin\theta$:

$(c + is)^4 = c^4 + 4ic^3s - 6c^2s^2 - 4ics^3 + s^4$ **M1**

Real parts: $\cos 4\theta = c^4 - 6c^2s^2 + s^4$ **A1**

$= c^4 - 6c^2(1 - c^2) + (1 - c^2)^2$ **M1**

$= c^4 - 6c^2 + 6c^4 + 1 - 2c^2 + c^4 = 8c^4 - 8c^2 + 1$ **A1 AG**

Second M1 for a binomial expansion with all five terms (powers of i need not yet be simplified). A1 for the real part with signs correct. Third M1 for replacing s^2 by $1 - c^2$ throughout. The final A1 needs the expansion of $(1 - c^2)^2$ seen.

(c) $1 - \sqrt{3}\,i = 2e^{-i\frac{\pi}{3}}$ **A1**

$\left(1 - \sqrt{3}\,i\right)^{10} = 2^{10}e^{-i\frac{10\pi}{3}} = 1024e^{i\frac{2\pi}{3}}$ **M1**

$= 1024\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -512 + 512\sqrt{3}\,i$ **A1**

A1 for both modulus 2 and argument $-\frac{\pi}{3}$ (accept $\frac{5\pi}{3}$). M1 for raising the modulus to the 10th power and multiplying the argument by 10. The GDC value $-512 + 886.81i$ alone earns A0M0A0: the question asks for de Moivre's theorem and the form $a + b\sqrt{3}\,i$; it is a useful check.

5. [Maximum mark: 13]

$w = -4\sqrt{2} + 4\sqrt{2}\,i$.

(a) $|w| = \sqrt{32+32} = 8$ **A1**

$\arg w = \dfrac{3\pi}{4}$, so $w = 8e^{i\frac{3\pi}{4}}$ **A1**

Accept 2.36 *for the argument.*

(b) $z = 8^{\frac{1}{3}} e^{i\frac{1}{3}\left(\frac{3\pi}{4}+2k\pi\right)}$ **M1**

$z_1 = 2e^{i\frac{\pi}{4}}$ **A1**

Successive roots differ in argument by $\dfrac{2\pi}{3}$ **M1**

$z_2 = 2e^{i\frac{11\pi}{12}}$ **A1**

$z_3 = 2e^{-i\frac{5\pi}{12}}$ **A1**

First M1 for the cube root of the modulus with the argument divided by 3 *(de Moivre with* $n = \frac{1}{3}$*).* z_3 *given as* $2e^{i\frac{19\pi}{12}}$ *(outside the stated range) earns A0 for that root only. FT from (a).*

(c) $2\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right) = \sqrt{2} + \sqrt{2}\,i$ **A1**

(d) *Method 1:* $z^3 - w = 0$ has no z^2 term **M1**

so the sum of the roots $= -\dfrac{0}{1} = 0$ **A1 AG**

Method 2: $z_1 + z_2 + z_3 = z_1\left(1 + e^{i\frac{2\pi}{3}} + e^{i\frac{4\pi}{3}}\right)$ **M1**

$= z_1\left(1 + \left(-\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i\right) + \left(-\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i\right)\right) = 0$ **A1 AG**

Accept adding the three roots in Cartesian form (exactly). A GDC sum alone, with no structure shown, earns M1A0.

(e) $w^{\frac{2}{3}} = \left(w^{\frac{1}{3}}\right)^2$, so square each cube root **M1**

$\left(2e^{i\frac{\pi}{4}}\right)^2 = 4e^{i\frac{\pi}{2}} = 4i$ **A1**

$\left(2e^{i\frac{11\pi}{12}}\right)^2 = 4e^{-i\frac{\pi}{6}} = 2\sqrt{3} - 2i, \quad \left(2e^{-i\frac{5\pi}{12}}\right)^2 = 4e^{-i\frac{5\pi}{6}} = -2\sqrt{3} - 2i$ **A1**

Accept $(w^2)^{\frac{1}{3}}$ *with* $w^2 = 64e^{i\frac{3\pi}{2}}$*. Accept* $\pm 3.46 - 2i$*. The final A1 needs both of the last two values. A single value* $(4i)$ *with no others earns M1A1A0. FT from (b).*